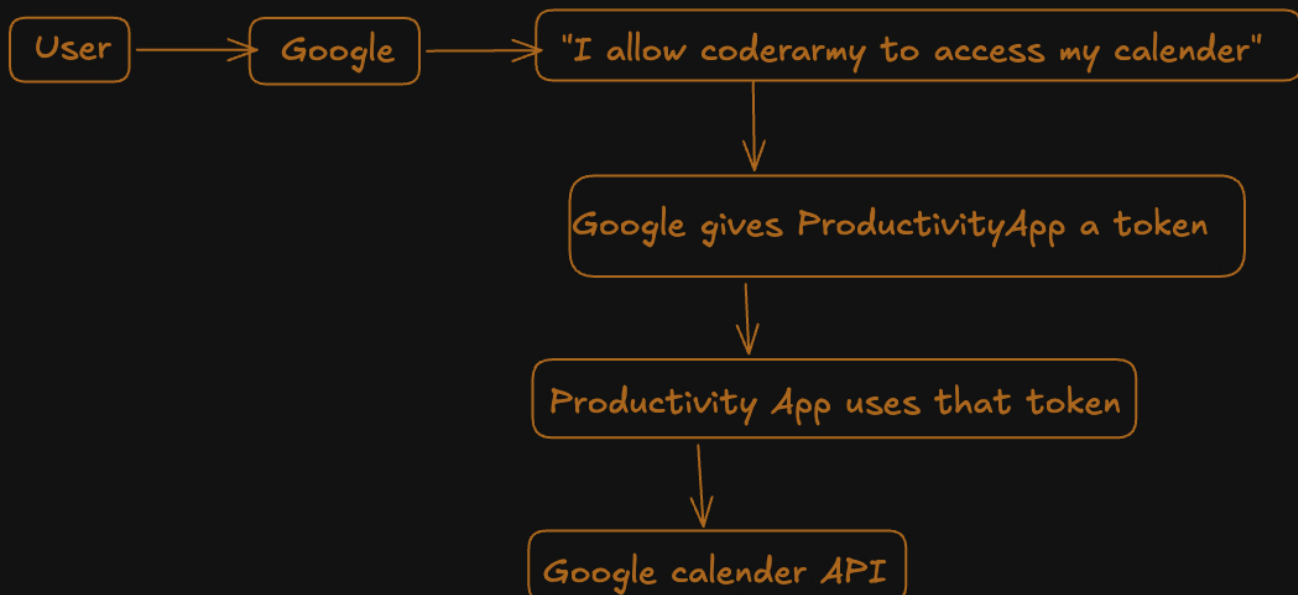
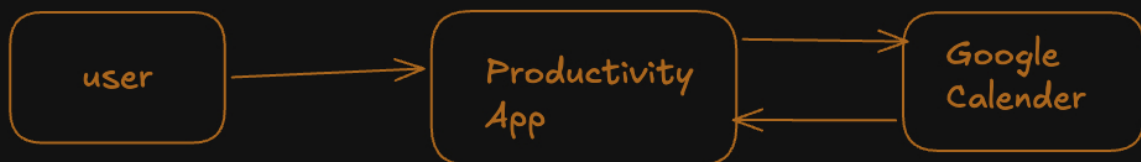
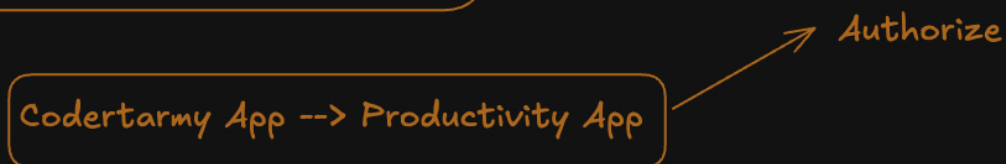
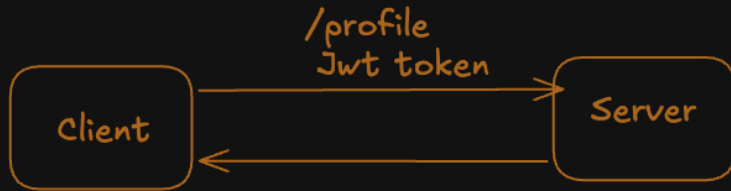
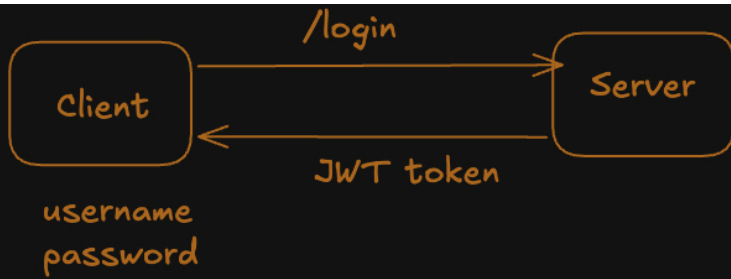




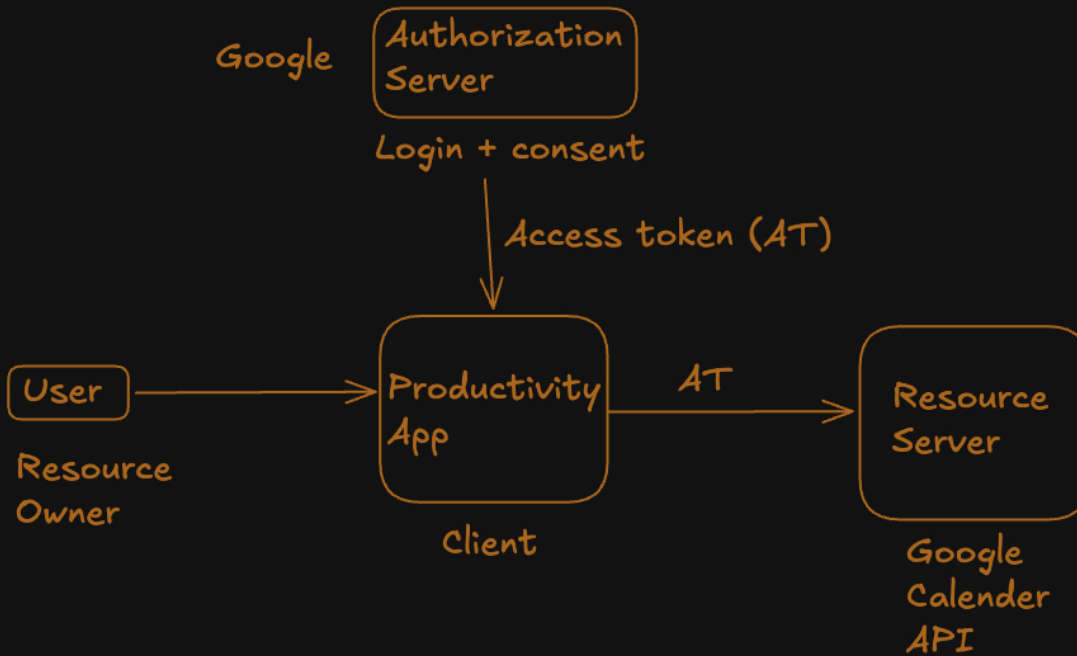
4 Keywords

Resource Owner --> User

Client —> Spring boot application

Authorization Server --> Google auth system

Resource Server --> Google Calender API



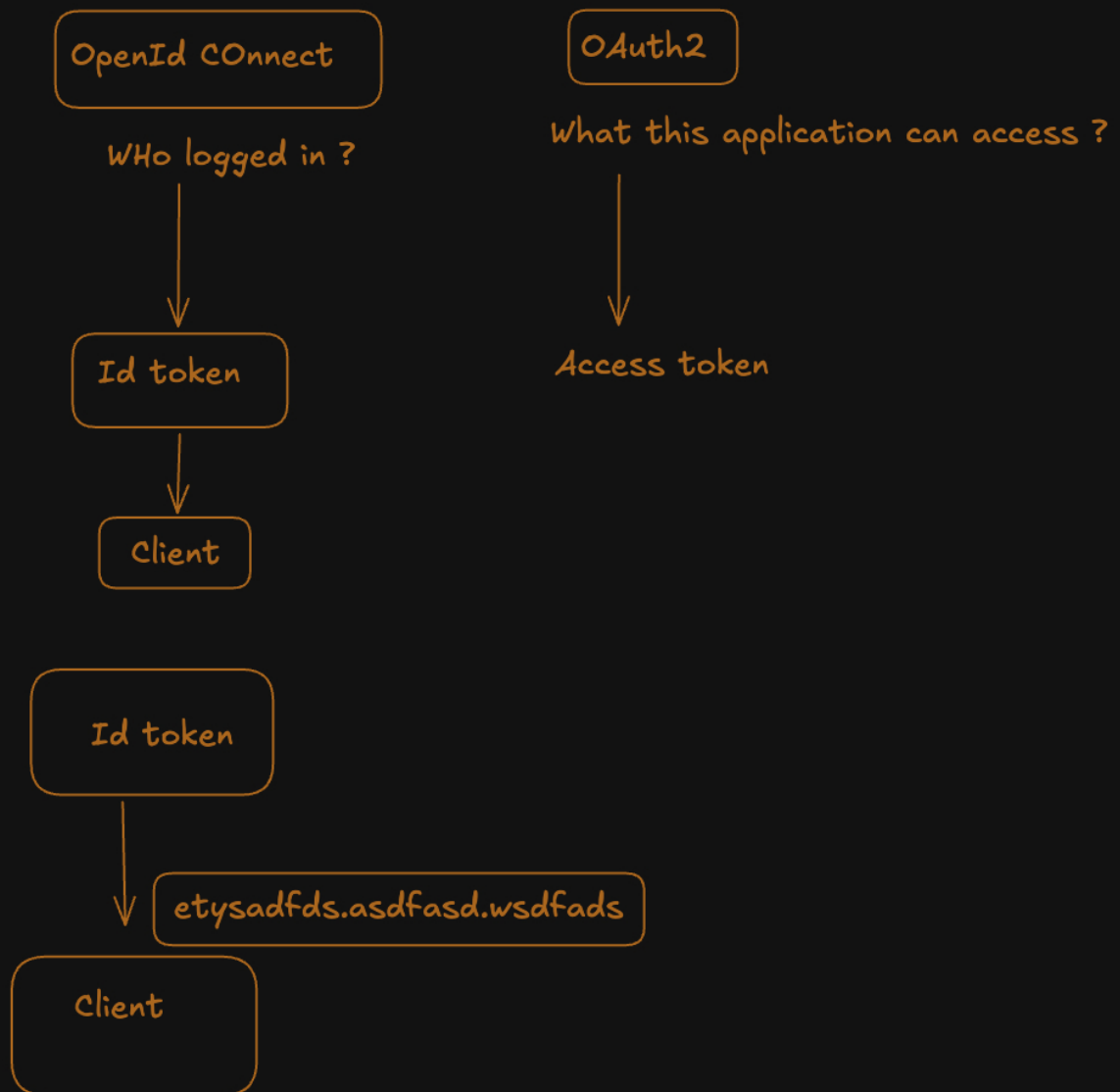
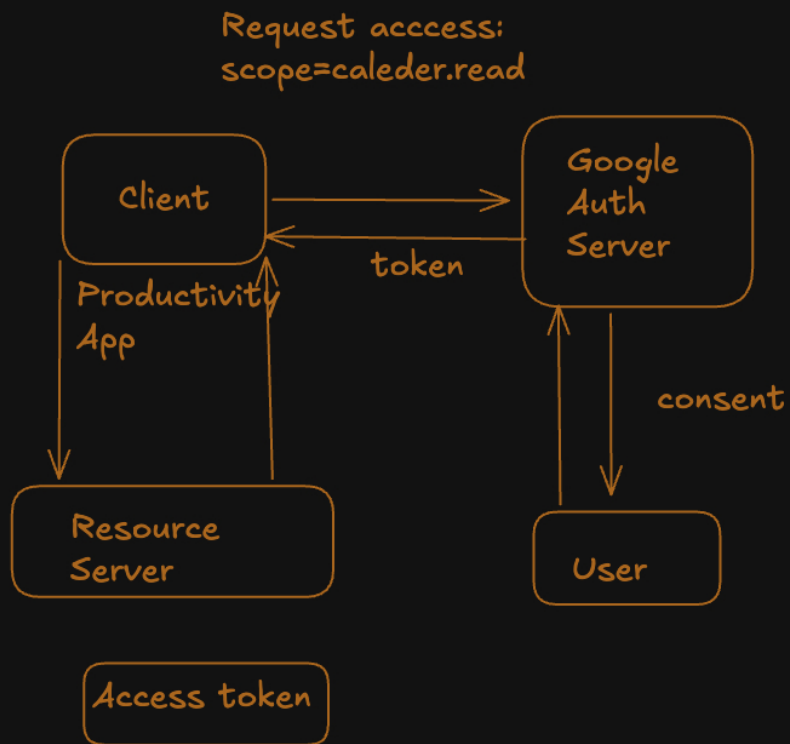
Scope and consent

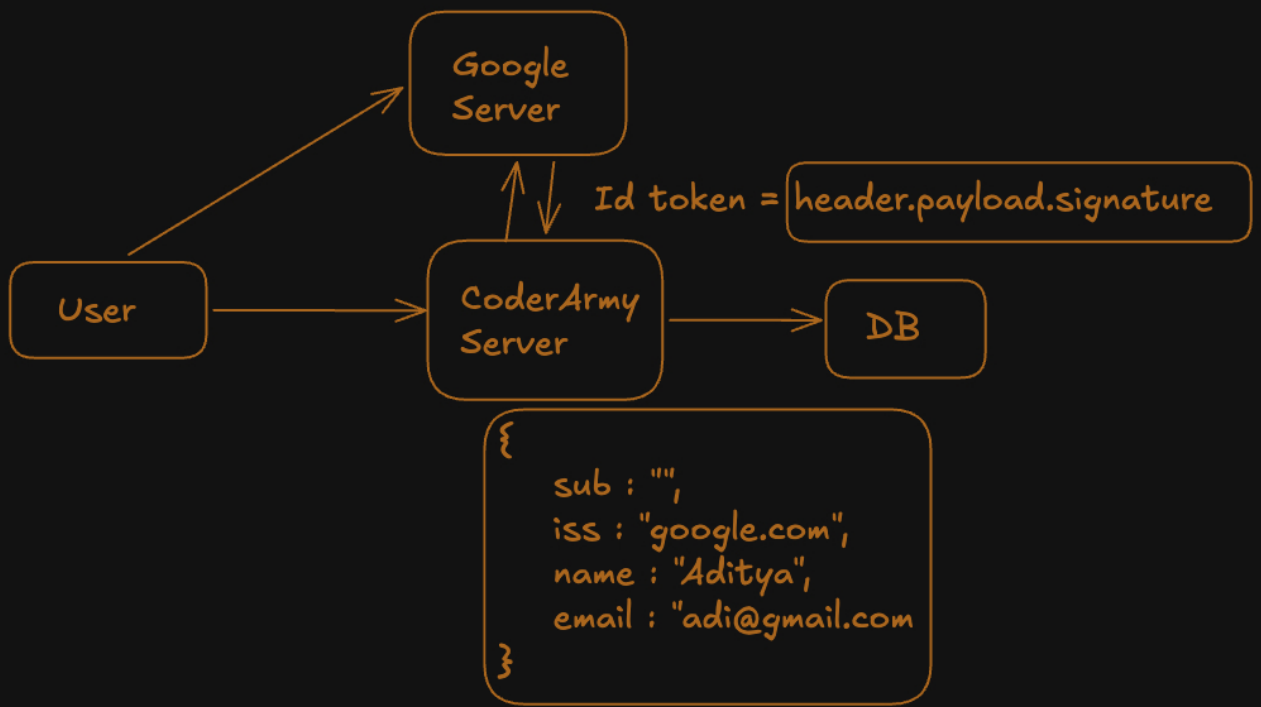
Scope --> calender.read

Consent :

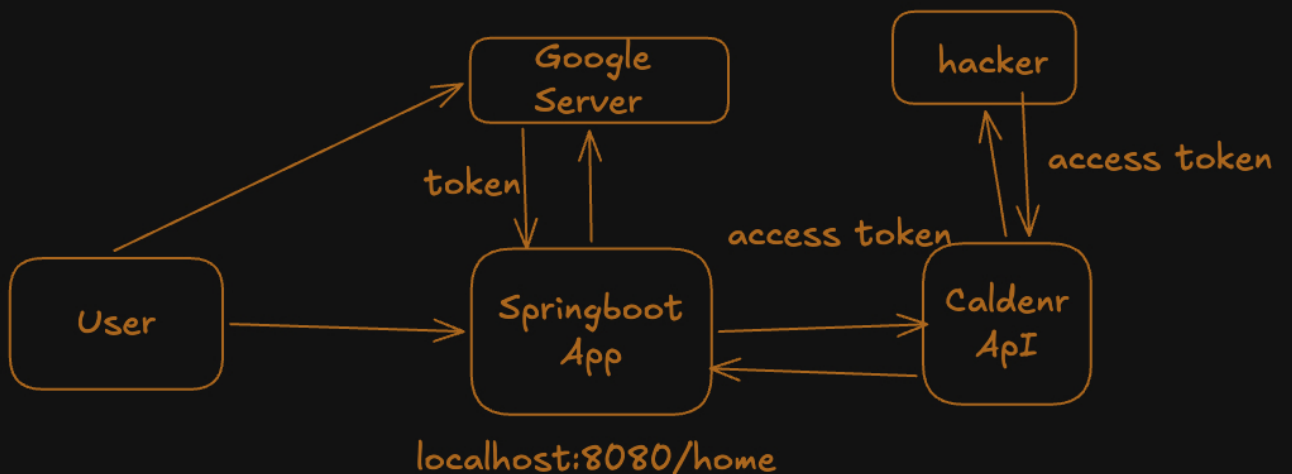
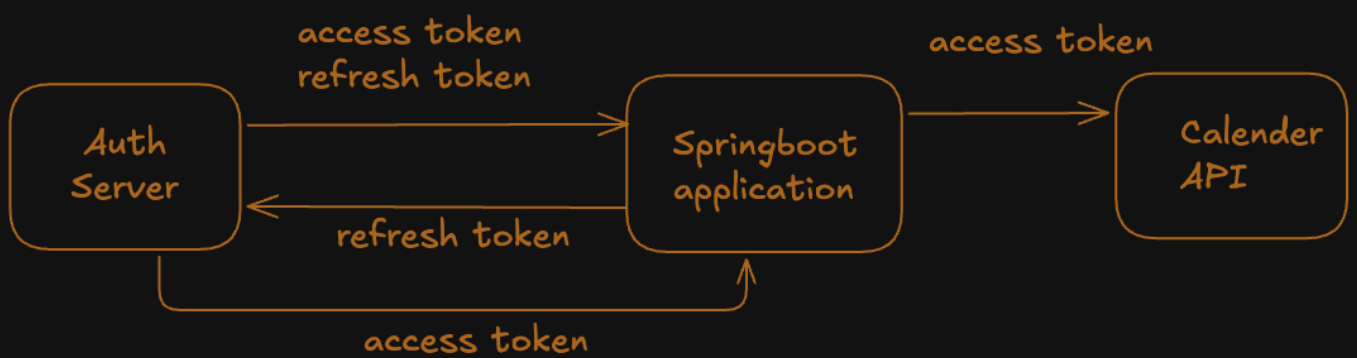
Productivity App wants permission to :

1. Read google calender
2. View your profile



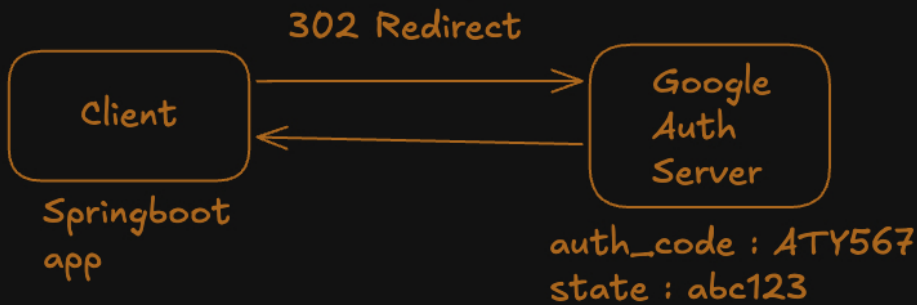


Access Token → Authorization token (Resource server)
Id Token → Authentication (Client)



Authorization code

1. Authorization Request



Auth Request :

clientId
redirection_uri
scope
response_type=code

state : random string

2. Google handles authentication

email
password

2FA

3. Authorization/ consent

4. Authorization code



localhost:8080/login/oauth2/code/google
?state=abc123&code=TYFGV456

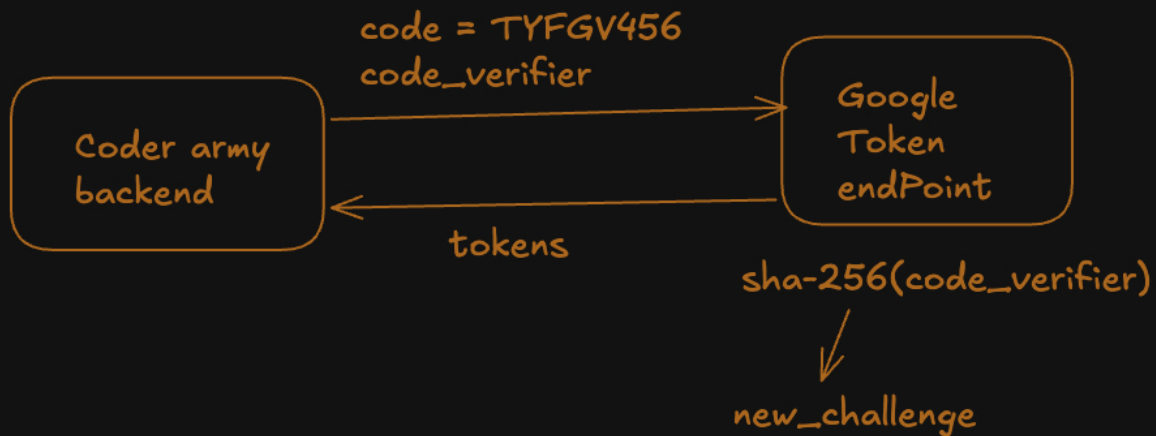
5. Exchange code to tokens



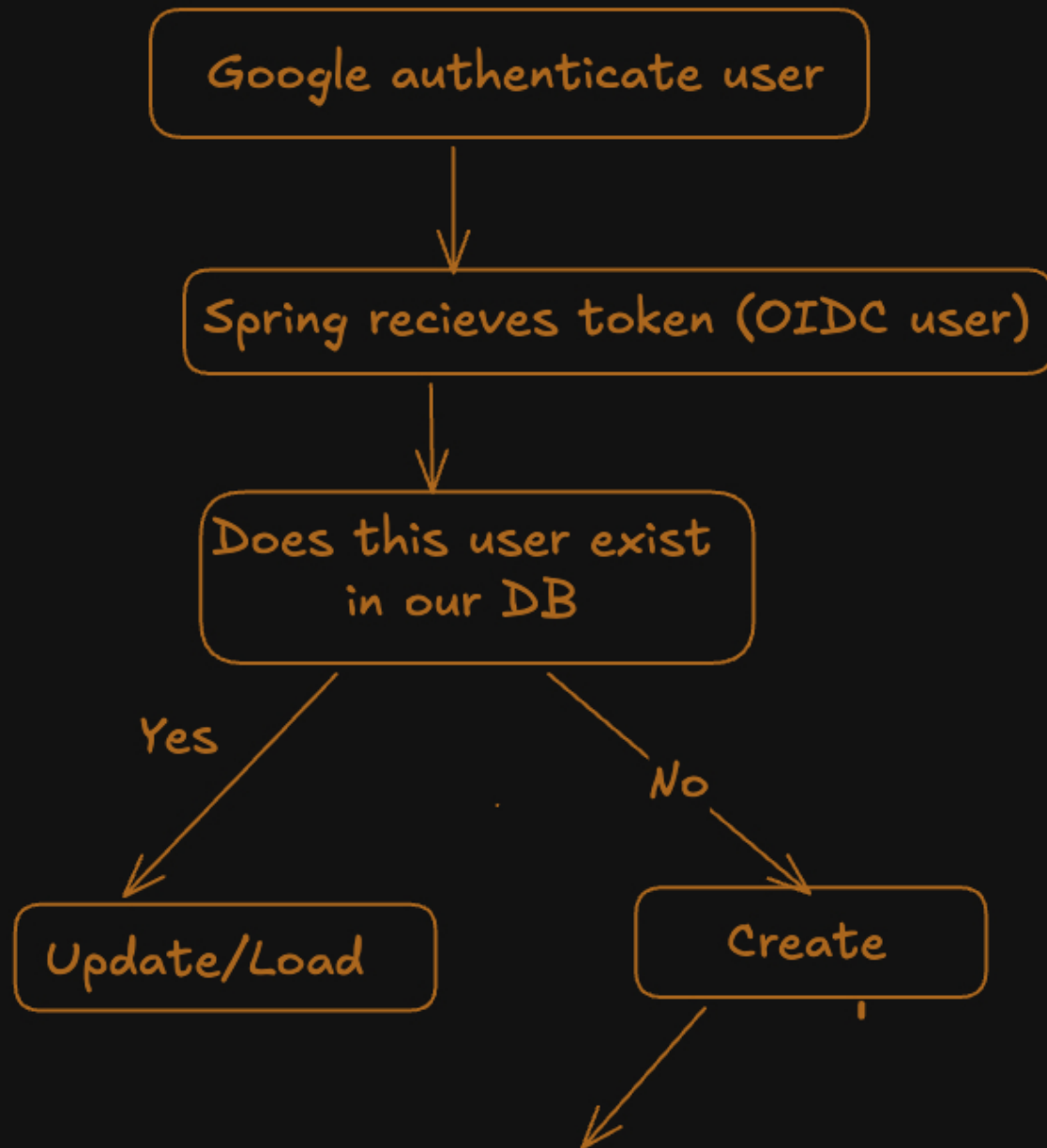
PKCE : Proof Key for Code Exchange



$\text{code_verifier} = \text{very-long-random-string}$
 $\text{code_challenge} = \text{SHA-256}(\text{code_verifier})$



clientId
clientSecret



```
{  
  sub : "109343242",  
  name :  
  email :  
}
```